Time Series Analysis for Proton Exchange Membrane Fuel Cell (PEMFC) Degradation Prediction

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Abstract

Estimating the remaining useful life (RUL) of proton exchange membrane fuel cells (PEMFCs) is crucial for reliable prognostics and durability enhancement. This study investigates and compares three recurrent neural network–based models, including vanilla recurrent neural networks (RNN), gated recurrent units (GRU), and long short-term memory (LSTM) networks, for long-term degradation modeling and RUL prediction of a five-cell PEMFC stack under dynamic operating conditions. The hyperparameters of all models are automatically optimized using the Optuna framework to minimize the root-mean-square error (RMSE) of stack voltage prediction. Model performance is evaluated using the RMSE, mean absolute error (MAE). The results indicate that gated recurrent architectures significantly outperform the conventional RNN in capturing long-term degradation dynamics. Among the evaluated models, the LSTM network demonstrates the highest prediction accuracy and robustness, achieving superior voltage prediction and RUL estimation performance compared to GRU and RNN models. These findings confirm the effectiveness of LSTM-based models for PEMFC health prognostics and long-term RUL prediction under realistic operating conditions.

1.Introduction

Proton exchange membrane fuel cells (PEMFCs) are promising zero-emission energy conversion systems for transportation and stationary power applications due to their high efficiency, fast start-up, and low noise. However, their widespread commercialization is hindered by limited operational lifetime and high maintenance costs, particularly under dynamic loading conditions. Accurate estimation of the remaining useful life (RUL) is therefore essential to enable predictive maintenance and improve system durability. Since PEMFC degradation is governed by complex and nonlinear dynamics, data-driven approaches have gained increasing attention. In this study, recurrent deep learning models, including recurrent neural networks (RNNs), gated recurrent units

(GRUs), and long short-term memory (LSTM) networks, are developed to model long-term voltage degradation and estimate RUL directly from operational data. Model hyperparameters are optimized using the Optuna framework to enhance prediction accuracy, and the proposed approach demonstrates strong capability for reliable degradation forecasting and online RUL estimation.

1.2.Deep Learning–Based Degradation Prediction Models

This section presents the architectures of the developed deep learning models used for degradation trend modeling and remaining useful life (RUL) estimation of PEMFC systems. The focus is on recurrent neural network–based architectures, which are particularly well suited for modeling sequential and temporal data.

1.2.1.Recurrent Neural Networks

Recurrent neural networks (RNNs) are designed to model fixed- and variable-length sequences by maintaining internal state variables that act as memory, enabling the learning of temporal dependencies in time-series data. However, conventional RNNs suffer from vanishing and exploding gradient problems, which limit their ability to capture long-term dependencies.

1.2.2.Long Short-Term Memory (LSTM) Networks

Long short-term memory (LSTM) networks are an enhanced RNN architecture developed to overcome the gradient instability issues of conventional RNNs. Owing to their capability to learn long-range temporal dependencies, LSTMs have been widely applied in speech recognition, natural language processing, and time-series forecasting.

Unlike standard RNNs, LSTMs consist of memory blocks rather than simple neurons, as illustrated in Fig. 1(a). Each memory block contains one or more memory cells regulated by three gating mechanisms: an input gate i_{ti}, a forget gate f_t, and an output gate t_o. These gates control the flow of information into, through, and out of the memory cell, allowing the network to selectively retain or discard information from previous time steps.

$$i_t = \sigma(W_{xi}\, x_t + W_{hi}\, h_{t-1} + b_i) \quad (1)$$

$$f_t = \sigma(W_{xf}\, x_t + W_{hf}\, h_{t-1} + b_f) \quad (2)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tanh(W_{xc}\, x_t + W_{hc}\, h_{t-1} + b_c) \quad (3)$$

$$o_t = \sigma(W_{xo}\, x_t + W_{ho}\, h_{t-1} + b_o) \quad (4)$$

$$h_t = o_t \odot \tanh(c_t) \quad (5)$$

1.2.3.Gated Recurrent Unit (GRU) Networks

The gated recurrent unit (GRU) is a simplified variant of the LSTM that reduces computational complexity while preserving the ability to capture long-term dependencies. In GRUs, the input and forget gates of the LSTM are merged into a single update gate, and the cell state is combined with the hidden state, as illustrated in Fig. 1(b). Additionally, a reset gate is introduced to control the contribution of past hidden states.

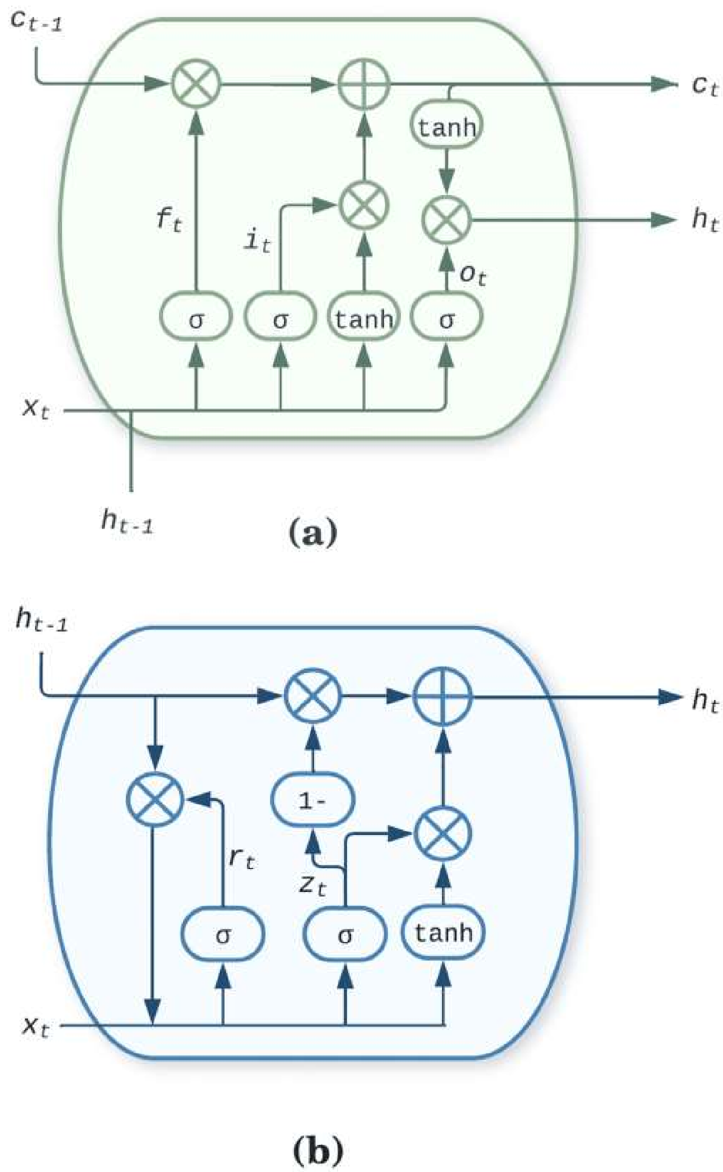


Fig1. (a) long short term memory; (b) gated recurrent unit

2.Dataset Description and Data Preprocessing

The experimental dataset used in this study was obtained from approximately seven-week-long degradation experiments conducted on two five-cell PEMFC stacks using the FR CNRS 3539 test bench at the FCLAB Research Federation, France. A comprehensive set of operational parameters was continuously monitored, including individual cell voltages, stack voltage, current, gas flow rates, temperature, and relative humidity, as summarized in Table 1.

Table 1. Measurements in the IEEE PHM 2014 Data Challenge Dataset

Model Input	Physical Significance
U1 – U5	Single-cell voltage (V)
Utot	Stack voltage (V)
I	Stack current (A)
J	Current density (A/cm^2)
TinH2	Inlet temperature of hydrogen (°C)
ToutH2	Outlet temperature of hydrogen (°C)
TinAIR	Inlet temperature of air (°C)
ToutAIR	Outlet temperature of air (°C)
TinWAT	Inlet temperature of cooling water (°C)
ToutWAT	Outlet temperature of cooling water (°C)
PinH2	Inlet pressure of hydrogen (millibar absolute, mbara)
PoutH2	Outlet pressure of hydrogen (mbara)
PinAIR	Inlet pressure of air (mbara)
PoutAIR	Outlet pressure of air (mbara)
DinH2	Inlet flow rate of hydrogen (L/min)
DoutH2	Outlet flow rate of hydrogen (L/min)
DinAIR	Inlet flow rate of air (L/min)
DoutAIR	Outlet flow rate of air (L/min)
DWAT	Flow rate of cooling water (L/min)
HrAIRFC	Estimated inlet air hygrometry (%)

2.1. PEMFC Dataset Description (FC1)

The first PEMFC stack, hereafter referred to as FC1, was operated under steady-state conditions at a constant load current of 70 A over a total operational duration of 1155 h. Each fuel cell within the stack had an active area of 100 cm^2. The FC1 dataset comprises 143,862 raw measurement samples, collected at a high temporal resolution. To ensure consistency with degradation analysis on an hourly time scale, the raw data were resampled to a 1 h resolution, resulting in 1155 time-indexed data points. The stack voltage profile of FC1 exhibits irregular and noisy behavior, which is characteristic of long-term PEMFC operation even under nominally steady loading conditions. Such fluctuations arise from complex interactions between electrochemical reactions, thermal effects, reactant distribution, and water management processes.

2.2. Data Preprocessing and Degradation Analysis

Prior to model training, comprehensive data preprocessing (Figure 2) was performed to improve signal quality and enhance model robustness. This included noise reduction and outlier mitigation, which are critical for accurately capturing long-term degradation trends. The availability of densely sampled measurements enabled effective smoothing, as sparse sampling is particularly sensitive to outliers and may lead to distorted degradation trajectories. To facilitate remaining useful life (RUL) estimation, a failure threshold was defined based on the stack voltage. This threshold represents the voltage level below which the PEMFC stack can no longer deliver sufficient power for reliable operation. The threshold serves as a quantitative criterion for degradation assessment and prognostics.

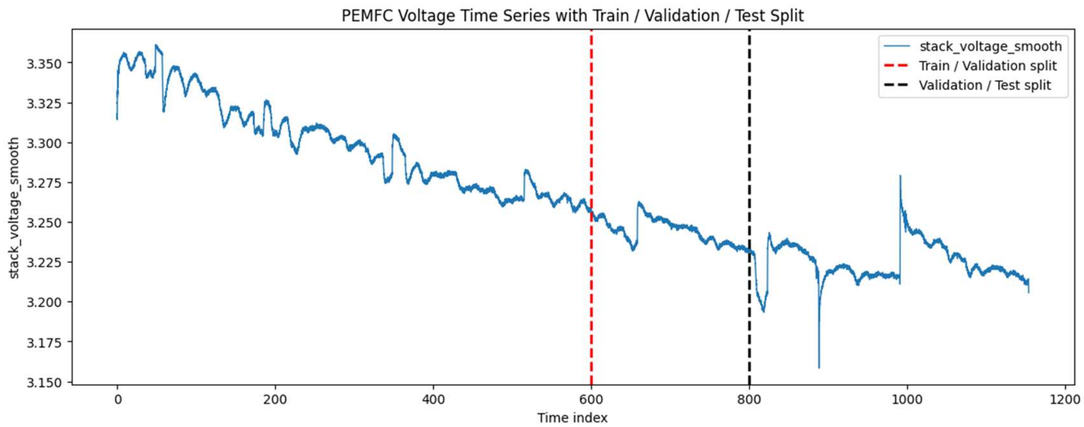


Fig. 2 (a) sampled, and split data for FC1

2.3.Failure Threshold Definition

The FC1 PEMFC stack was operated under steady-state conditions for 1155 hours. Although the operating current was constant, the measured voltage signal still contained noise and small fluctuations. To accurately capture the long-term degradation trend, the raw voltage data were first smoothed using a Savitzky–Golay filter (Figure3). This filter is commonly used in PEMFC prognostics because it effectively reduces noise while preserving the underlying degradation slope. After smoothing, a failure threshold was defined as 96% of the initial stack voltage at the start of the experiment. This threshold is consistent with recent PEMFC prognostics studies and represents the voltage level below which the stack can no longer deliver reliable power for safe operation. For FC1, this resulted in a failure threshold of 3.2028 V. It is important to note that the choice of Savitzky–Golay filter parameters, such as window length and polynomial order, influences the smoothness of the voltage profile and therefore affects the resulting failure threshold. Careful selection of these parameters is essential to ensure physically meaningful RUL estimation. To enable supervised deep learning, the time-series data were reformulated into a supervised learning problem by introducing time-lagged input–output pairs. A one-step time lag was applied to all measurements, meaning that each time step was used to predict the next step. This transformation produced a supervised dataset of 1154 hourly samples for FC1, which is one hour less than the original data due to the time lag. This setup allows recurrent neural networks such as LSTM, GRU, and RNN to learn temporal dependencies and predict the remaining useful life of the fuel cell stack.

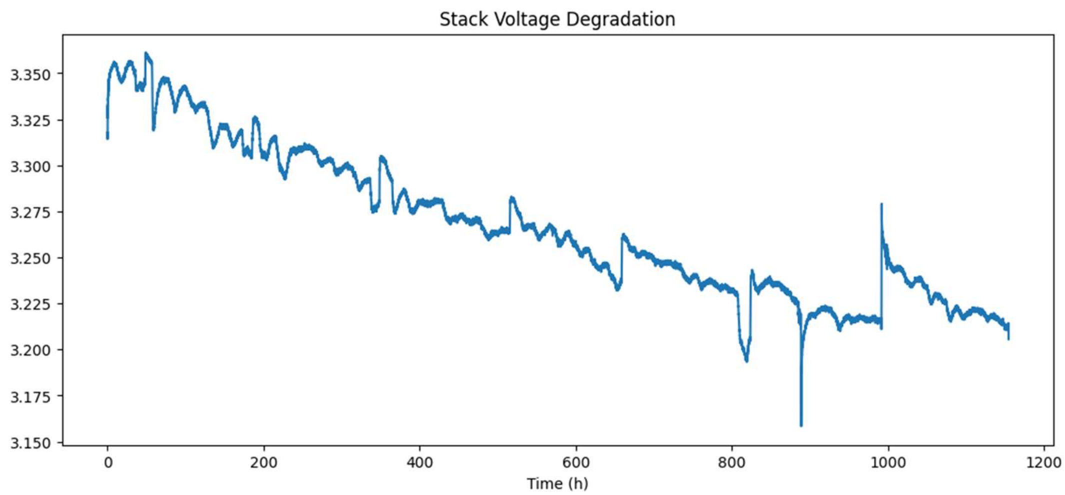


Fig3. Savitzky–Golay filter for Stack voltage

The training, validation, and test sets were normalized using min-max normalization. Performing normalization before splitting the dataset can lead to data leakage, as statistical information (e.g., mean and variance) from the training set may inadvertently influence the validation and test sets. This can introduce bias in model performance evaluation when the models are exposed to unseen data. To prevent this, the datasets were normalized after splitting into training, validation, and test sets. All features and the stack voltage in the training set were scaled to the [0, 1] range using min-max normalization.

$$\overline{x} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

The predictive performance of each model was evaluated using four metrics for long-term degradation prediction: the mean absolute square (MAE), and root-mean-square error (RMSE). Overfitting during training was mitigated using dropout regularization and an early stopping strategy. Overfitting occurs when a model achieves high accuracy on the training set but performs poorly on the test set, as it memorizes rather than generalizes from the training data. Dropout prevents overfitting by randomly deactivating neurons and their connections during training. In early stopping, training is halted when the validation loss does not improve for a predefined number of epochs, known as the patience value. The patience value defines how many consecutive epochs without improvement are tolerated before stopping. Dropout was treated as a tunable hyperparameter, and all models were trained for 150 epochs with early stopping using a patience value of 10.

To achieve fast convergence of prediction results, three gradient-based neural network optimizers were evaluated for their predictive performance: stochastic gradient descent (SGD), and Adam. Each model was trained over 250 trials using an early stopping strategy with a patience value of 30. For each model, the dropout rate was optimized within the range [0.01, 0.2], while the optimizer was selected from the set {SGD, Adam}. The learning rate for both the SGD and Adam optimizers was searched over a logarithmically spaced range of [1×10^{-5}, 1×10^{-1}]. For SGD, the momentum parameter was tuned over a logarithmically spaced range of [1×10^{-3}, 0.99].

3.EDA (Exploratory Data Analysis)

3.1.Distribution of Stack Voltage in a PEMFC Stack

Figure 4 shows the distribution of stack voltage (V) in a PEMFC stack, presented as a histogram that captures how frequently different voltage levels occur during operation. The x-axis represents the measured stack voltage, spanning approximately 3.10–3.35 V, while the y-axis denotes the frequency of occurrence of each voltage range. The dominant peak in the distribution corresponds to the most common operating voltage of the stack under the given conditions, representing the typical electrochemical working point. A narrow, well-centered distribution around this peak indicates stable operation with effective control of load, temperature, and reactant supply. In contrast, a wider spread reflects increased voltage variability, which may arise from dynamic load changes, evolving cell-to-cell imbalance, or aging-related effects.

The shape of the distribution provides further insight into stack health. A near-Gaussian profile suggests random fluctuations around a steady operating state, whereas skewness or a systematic shift of the peak over time may indicate gradual performance drift associated with degradation. The presence of multiple peaks would imply transitions between distinct operating regimes, such as idle and high-load conditions.

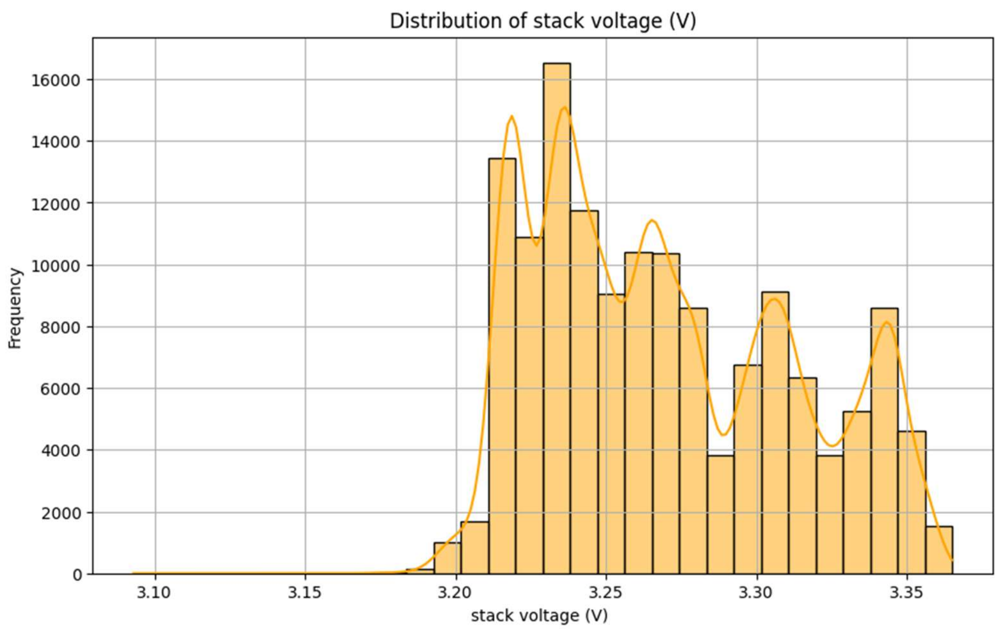


Fig 4. the distribution of stack voltage (V) in a PEMFC stack

3.2. PEMFC Stack Voltage Degradation Over Time

This plot 6 illustrates the voltage decay of a PEMFC stack over prolonged operation (up to 1200 hours). The x-axis shows operating time (h)**,** while the y-axis represents the stack voltage (V)**,** typically ranging from 3.35 V at the start to around 3.10 V at the end of life.

At the beginning, the stack operates near its maximum voltage, reflecting well-conditioned membranes, healthy catalysts, and uniform reactant distribution. Over time, the voltage gradually decreases due to membrane drying or flooding, catalyst degradation, carbon corrosion, and electrode contamination**.** The curve typically shows an initial break-in drop, followed by a steady, slow decline**,** with potential accelerated drops near end-of-life**.** Temporary recoveries may appear due to improved hydration or transient operating conditions.

From an engineering perspective, the slope of this degradation curve quantifies the stack's durability and informs RUL estimation, maintenance planning, and operating strategies**.** Stepwise drops or sudden dips may indicate localized failures or system events, while a smooth decline signals normal aging. Overall, this visualization provides a clear, quantitative view of PEMFC stack performance and long-term reliability.

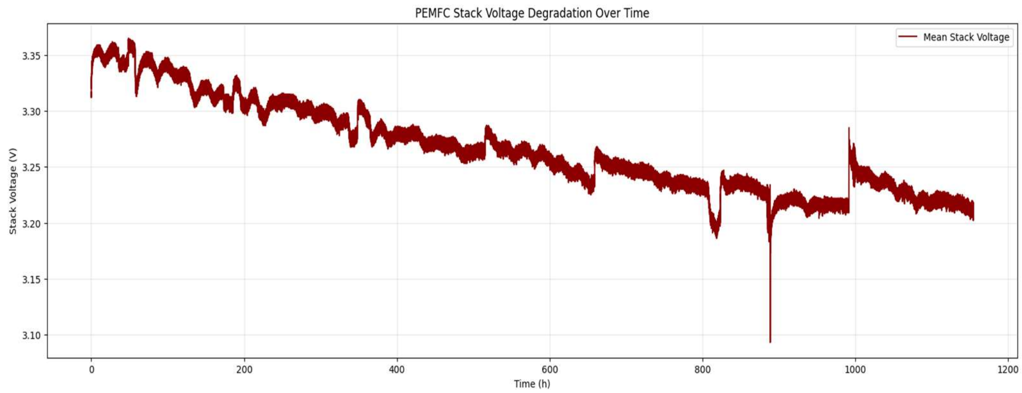


Fig 6. illustrates the voltage decay of a PEMFC stack over prolonged operation (up to 1200 hours).

3.3.Cell Voltage Imbalance Over Time in a PEMFC Stack

Figure 7 illustrates the evolution of cell voltage imbalance in a proton exchange membrane fuel cell (PEMFC) stack over up to 1200 hours of operation. The imbalance is quantified by the standard deviation of individual cell voltages, serving as a sensitive indicator of stack-level degradation. The x-axis represents cumulative operating time (h), while the y-axis shows the cell voltage standard deviation (V), which reflects the degree of non-uniform electrochemical performance among cells. Low values indicate a well-balanced stack, whereas increasing values signify growing cell-to-cell heterogeneity.

At the beginning of operation, the stack exhibits minimal voltage dispersion (≈0.002–0.003 V), consistent with uniform membrane hydration, balanced reactant distribution, and healthy catalyst layers. With continued operation, the voltage standard deviation gradually increases, potentially reaching ~0.02–0.025 V after prolonged aging (~1200 h). This trend is attributed to non-uniform degradation mechanisms such as localized membrane dehydration or thinning, spatially heterogeneous catalyst degradation and ECSA loss, and variations in gas transport, water management, and thermal conditions.

The progressive increase in voltage imbalance is a critical degradation marker in PEMFC stacks, as lower-voltage cells experience higher overpotentials that accelerate local degradation and increase the risk of premature failure. Consequently, cell voltage standard deviation provides early insight into internal stack health and is highly relevant for degradation tracking, Remaining Useful Life (RUL) prediction, and prognostics and health management (PHM) of PEMFC systems.

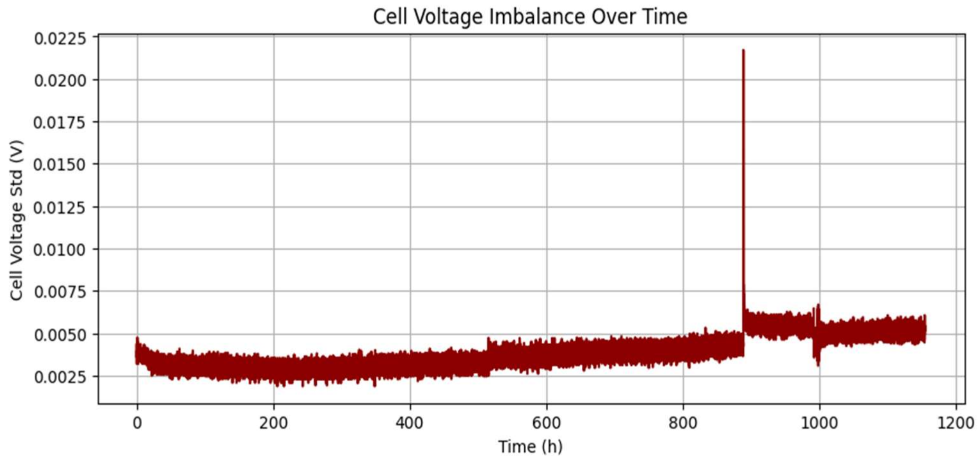


Fig. 7.Cell Voltage Imbalance Evolution in PEMFC Stack (0–1200 hours)

3.4.Impact of Thermal Stress on PEMFC Stack Voltage

This plot 8 illustrates the relationship between thermal stress (ΔT Air, °C) and stack voltage (V) in a PEMFC system. The x-axis represents the temperature difference across the stack or between inlet and outlet air, while the y-axis shows the stack voltage under operating conditions. As ΔT increases, representing greater thermal gradients or uneven heating, the stack voltage generally decreases. Low ΔT corresponds to uniform temperature distribution and stable voltage, whereas high ΔT introduces hotspots, accelerates membrane drying/flooding, reduces catalyst activity, and increases internal resistance, all contributing to voltage drop.

The observed trend is a negative correlation: as thermal stress rises, stack performance declines. The curve may be nonlinear, showing minimal effect at small ΔT and sharper voltage loss at higher ΔT. Data clusters may also indicate different operating conditions, such as load or humidity variations. Engineering Implications**:** This plot emphasizes the importance of thermal management, guiding airflow design, cooling strategies, and operational limits. Monitoring ΔT helps predict long-term stack degradation and supports durability optimization, ensuring reliable PEMFC performance under real-world conditions.

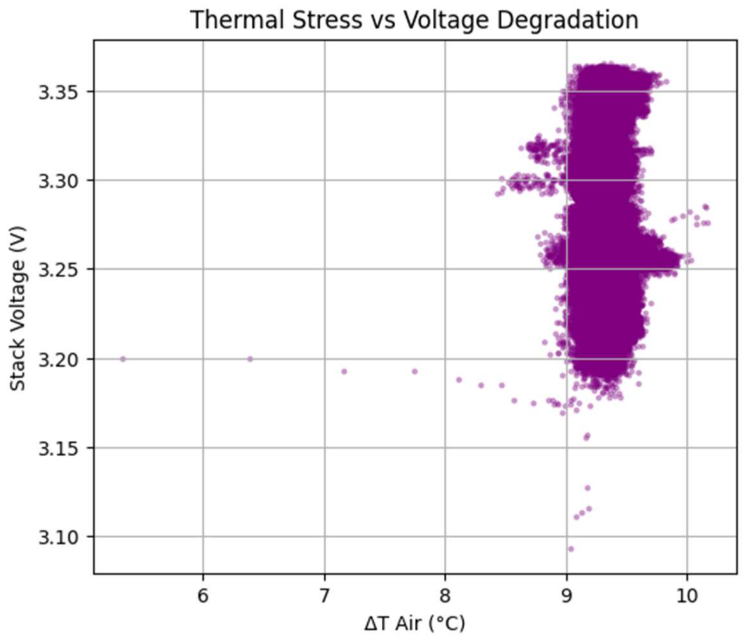


Fig. 8. Relationship Between Thermal Stress (ΔT Air, °C) and Stack Voltage (V)

Figure 9 shows the evolution of stack and individual cell voltages over 1200 hours of PEMFC operation. The y-axis (0–3.5 V) captures both the overall stack voltage (~3.0–3.5 V) and individual cell voltages (~0.6–0.8 V).

- Trends: Both stack and cell voltages stability over time
- Insights: The gap between stack and cell voltages indicates weak cells limiting overall performance.
- Implications: Monitoring both stack and cell voltages enables RUL prediction, early detection of failing cells, and assessment of cell balancing effectiveness.

This plot provides a concise view of PEMFC degradation, highlighting how individual cell behavior drives overall stack health and informs maintenance strategies.

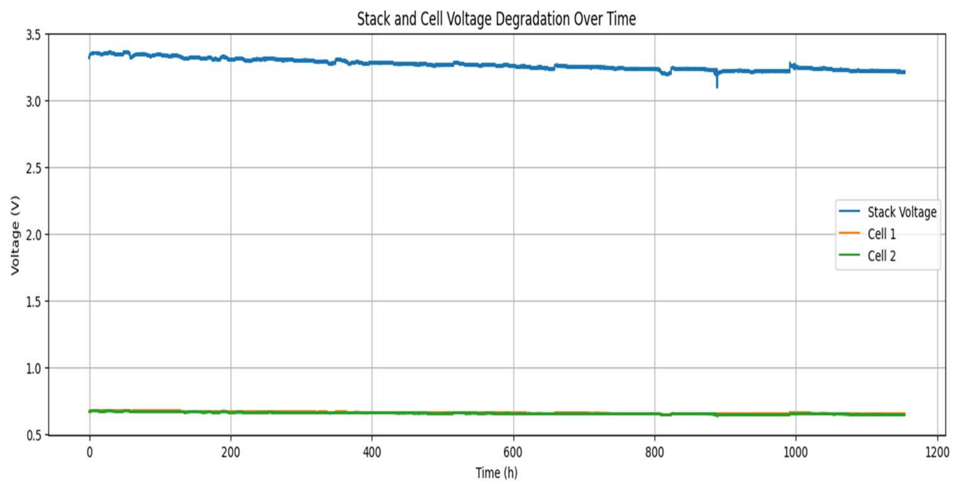


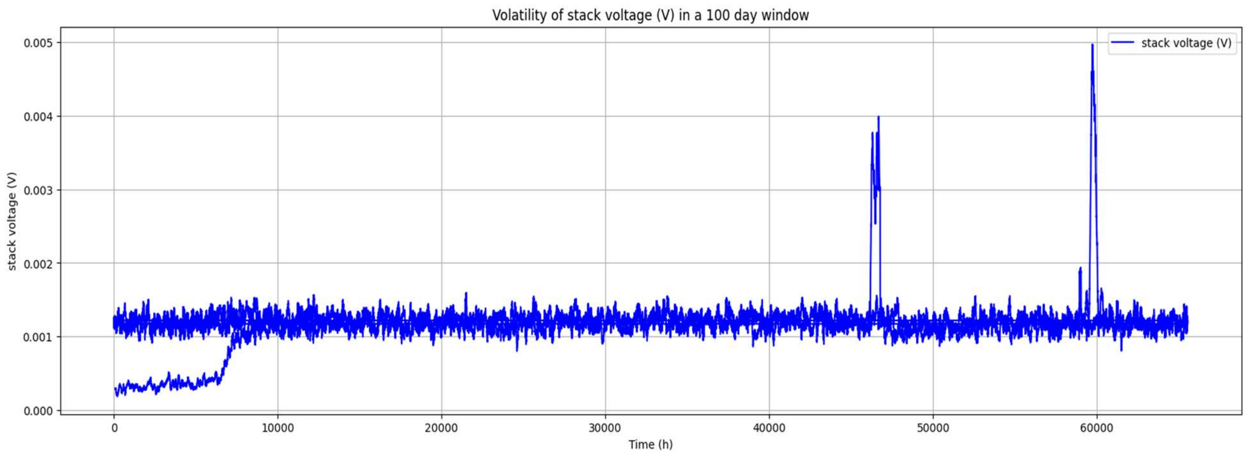
Fig. 9. Evolution of Stack and Individual Cell Voltages over 1200 Hours

3.5. Stack Voltage Volatility Over Time in PEMFCs

Figure X presents the stack voltage of a PEMFC system over a 100 equivalent in operating hours, highlighting short-term fluctuations and operational stability.

- **Axes:**
    - X-axis: Time (h) – cumulative operating hours
    - Y-axis: Stack Voltage (V)
- **Key Observations:**
    - Volatility captures short-term ups and downs, load cycles, and spikes/dips caused by operational events.
    - Gradual long-term drift indicates underlying degradation, while a stable mean with minimal fluctuations reflects consistent stack performance.
    - High voltage volatility may indicate thermal management issues, reactant flow instability, or control system/sensor anomalies.
- **Practical Significance:**
    - Monitors PEMFC operational stability in real time.
    - Provides early warning of potential performance issues before significant degradation occurs.
    - Supports maintenance planning and operational optimization.

This plot quantifies both short-term variability and long-term trends in PEMFC stack voltage, offering insights into system reliability, control effectiveness, and early degradation signals.



4. Hyperparameter Evaluation and Model Optimization

The optimal hyperparameter configurations, identified using OPTUNA with Hyperband pruning over 500 search trials within the defined hyperparameter space, are presented for the FC1 and FC2 datasets in Tables 2.

Table 2 – Optimal hyperparameter values for models trained on the FC1 dataset.

4.1.Data Preprocessing and Dataset Formation

The FC1 dataset was down-sampled to hourly intervals and smoothed using a Savitzky–Golay filter to remove noise while preserving the degradation trend. A sliding window length of 21 was used for smoothing to capture long-term trends effectively. The time-series data were reformulated into a supervised learning problem by introducing a one-step time lag, meaning that measurements at time $t-1$ were used to predict the stack voltage at time t. This resulted in a supervised dataset consisting of 24 input features (all stack measurements at $t-1$) and one target variable (stack voltage at t). The dataset was then split into training, validation, and test sets in a 50:10:40 ratio, and normalized using min-max scaling. Importantly, the min and max values were calculated only from the training set to prevent data leakage.

4.2.Optimizer Comparison and Learning Rate Impact

To compare optimizers, the same LSTM architecture was trained using Adam and SGD, both with a fixed learning rate of 0.003. The training history revealed how each optimizer affected convergence:

- Adam typically showed faster convergence and lower validation loss due to its adaptive learning rate mechanism.
- SGD often required more epochs to converge and was more sensitive to the learning rate and momentum settings.

In addition, multiple learning rates were evaluated to study their effect on training dynamics. Models were trained using a range of learning rates, and the training history for each rate was recorded. This analysis confirmed that a learning rate of 0.003 provided a good balance between convergence speed and stability.

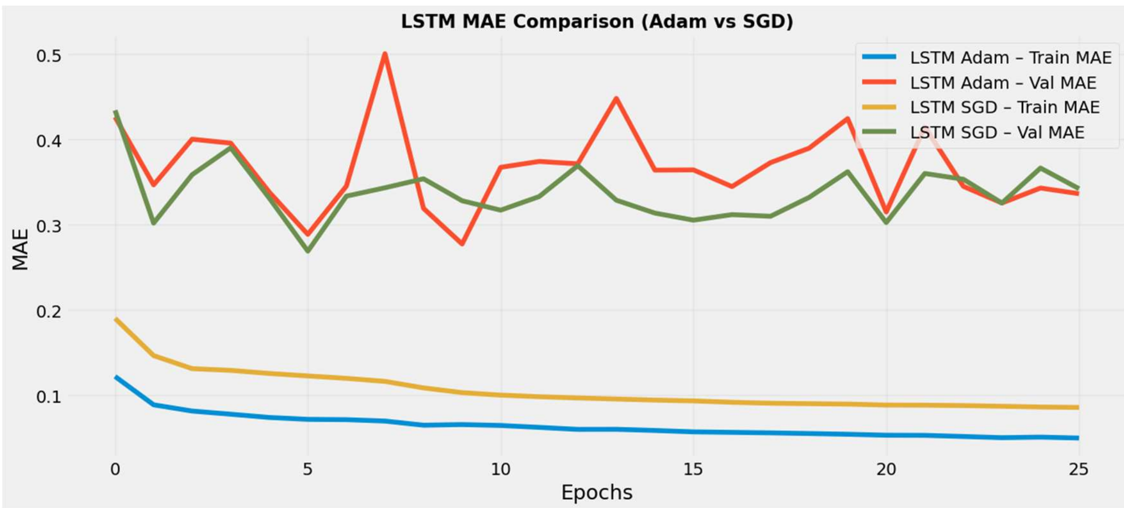


Fig10. LSTM validation vs optimizer

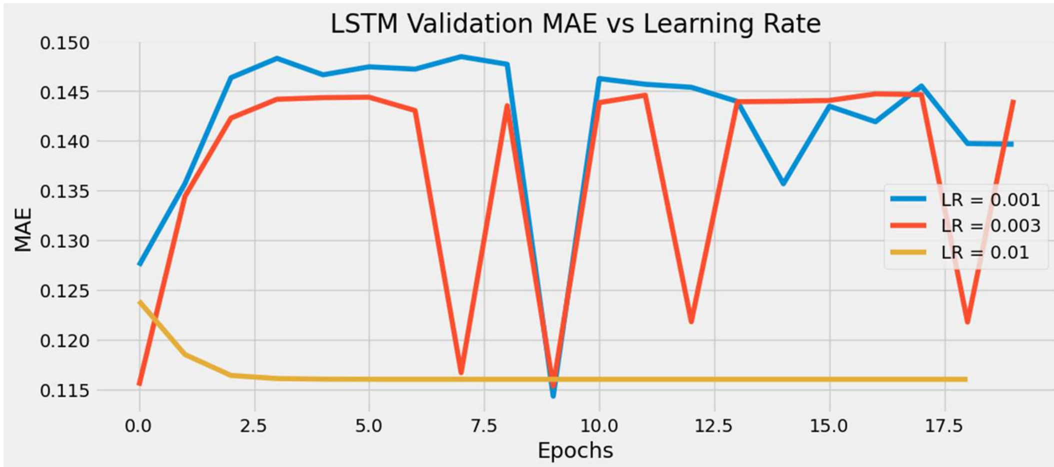


Fig11. LSTM validation learning rate

4.3.Effect of Dropout on Model Performance

Two LSTM models were compared: one without dropout and one with dropout layers added between LSTM blocks. The dropout rate was set to 0.2 in the dropout model.

- Without dropout, the model tended to fit the training data more closely, which sometimes led to overfitting.
- With dropout, the model showed improved generalization, resulting in smoother predictions and better performance on the validation set.

This comparison highlighted that dropout is beneficial for preventing overfitting in recurrent neural networks, especially when predicting noisy PEMFC voltage signals.

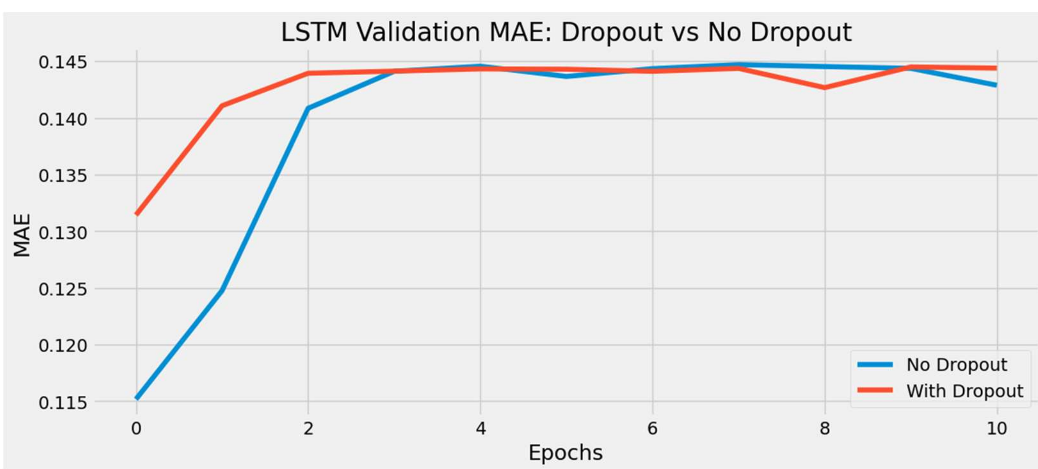


Fig12. LSTM validation dropout and No dropout

The hyperparameter evaluation confirmed that the combination of SGD, and Adam along with dropout regularization and carefully tuned learning rates, can significantly affect the performance of LSTM-based prognostics. The optimization strategy effectively improved convergence speed and prediction accuracy, while early stopping ensured model generalization. Ultimately, the selected model configuration was able to learn the degradation trend of the FC1 stack voltage and provided reliable predictions for RUL estimation. Interestingly, for all models, the best-performing optimizer for both the FC1 datasets was found to be Adam, along with the most suitable activation function.

5. Prediction Results: Classical Time-Series Models (ARIMA, SARIMA, Prophet)

The predictive performance of classical statistical time-series models—ARIMA, SARIMA, and Prophet—was evaluated on the FC1 and FC2 datasets to benchmark their capability in capturing long-term stack voltage degradation and estimating remaining useful life (RUL). These models were trained using the same preprocessed and normalized datasets as the deep learning models to ensure a fair comparison.

5.1.ARIMA Model Performance

The ARIMA(2,1,2) model effectively captures the global degradation trend of the FC1 stack voltage, as evidenced by the close overlap between observed and fitted values as shown in Figure 13. The smooth downward trajectory indicates strong performance in modeling long-term, persistent voltage decay, which is characteristic of fuel cell aging.

However, the diagnostic plots reveal clear limitations in short-term dynamics. Residual autocorrelation (ACF/PACF) and heteroskedasticity indicate that the model fails to fully represent transient operating effects, including load variations, abrupt voltage drops, and accelerated degradation near end-of-life. These behaviors violate ARIMA's linear and constant-variance assumptions. Although the quantitative errors (MAE = 0.0112, RMSE = 0.0134) appear small, they become non-negligible in a prognostics context**,** particularly during the final degradation phase where nonlinear mechanisms dominate. The near-perfect in-sample fit further suggests over-smoothing**,** limiting the model's generalization capability.

In summary**,** the diagnostic plots confirm that ARIMA provides a reliable baseline trend estimator**,** but its linear structure restricts its ability to capture complex degradation dynamics. This explains the combination of strong mean trend tracking and a poorly behaved residual structure observed in the diagnostics, highlighting the need for nonlinear or hybrid models for accurate RUL prediction.

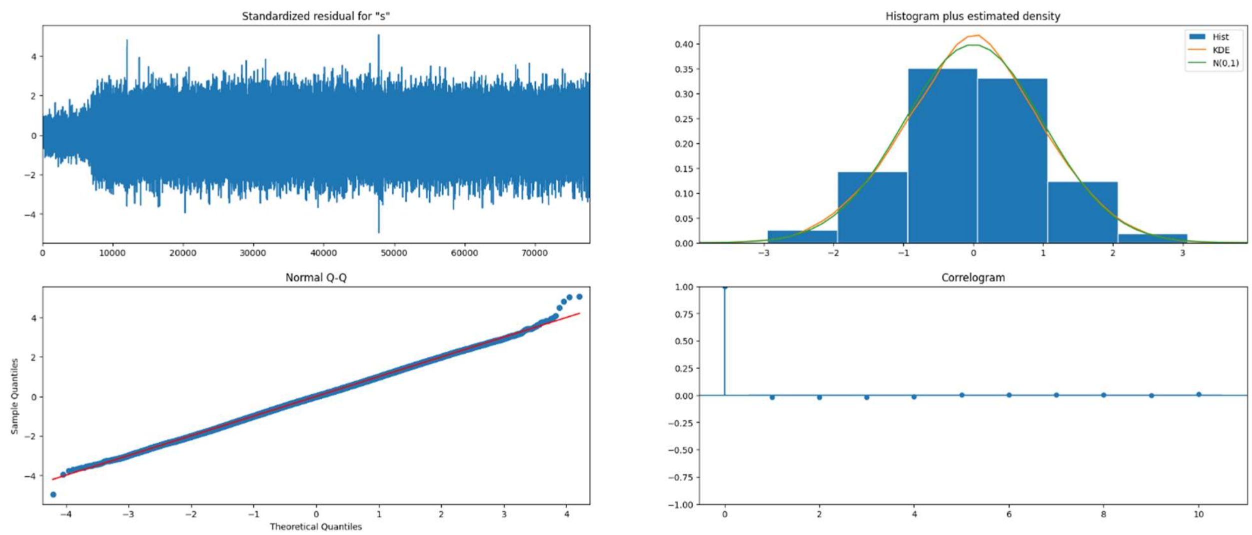


Fig. 13. ARIMA Trend Fit and Diagnostic Plots

5.2.SARIMA Model Performance

When we examined the PEMFC stack voltage over its operational lifetime, the data showed a clear, long-term degradation trend punctuated by small short-term fluctuations as shown in Figure 14. To capture this behavior, we applied a SARIMAX(1,1,1) model, fitting it to the concatenated training and validation data. The model coefficients reveal the underlying dynamics: the AR(1) term of 0.873 indicates that the voltage at the previous timestep is a strong predictor of the current voltage—highlighting the memory effect in fuel cell degradation. The negative MA(1) coefficient of -0.728 demonstrates that the model corrects past errors in the opposite direction, effectively smoothing transient fluctuations. The very small residual variance (`sigma² ≈ 1.38e-08`) reflects the model's ability to track the mean degradation trend with high precision.

Diagnostic analysis tells a complementary story. While the residuals are roughly normally distributed and symmetric, they exhibit heteroskedasticity and slight autocorrelation (Ljung-Box Q $p < 0.001$, Heteroskedasticity $p < 0.001$). This suggests that although the model captures the overall trend remarkably well, short-term voltage spikes, operational variability, and nonlinear electrochemical effects are not fully captured.

Visually, the diagnostic plots reinforce this narrative: standardized residuals fluctuate over time, occasionally exhibiting periods of higher volatility, while the histogram and Q-Q plots confirm near-normality of errors. The strong AR term ensures that the model faithfully follows the long-term decline, but the residual patterns remind us that PEMFC degradation is influenced by processes beyond simple linear memory, including temperature-dependent kinetics, hydration cycles, and membrane aging. In practical terms, the SARIMAX(1,1,1) model offers a reliable baseline for forecasting stack voltage and estimating Remaining Useful Life (RUL). It excels in modeling the smooth downward trend, providing interpretable, data-driven predictions. However, for precise short-term forecasting or capturing complex electrochemical phenomena, additional modeling strategies**,** such as hybrid models combining linear SARIMAX with machine learning or nonlinear approaches, may be required.

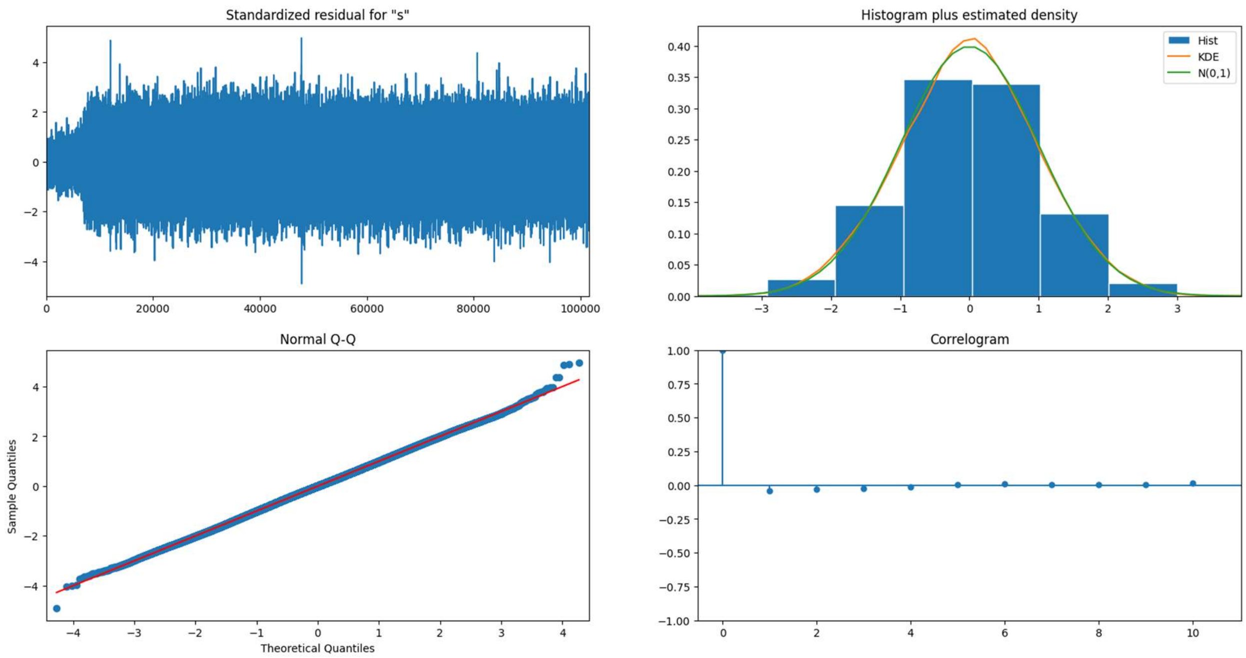


Fig. 14. SARIMA Trend Fit and Diagnostic Plots

5.3.SARIMAX Forecast Overview – PEMFC Stack Voltage

The plot 15 shows the actual stack voltage (green), the SARIMAX forecast (red dashed line), and the 95% confidence interval (pink shaded area). Over the 350 test points, the model captures the overall downward degradation trend of the stack voltage. Most actual values lie within the confidence bounds, indicating reasonable uncertainty quantification. The confidence interval widens slightly over time, reflecting increasing prediction uncertainty, while short-term fluctuations are less accurately tracked, consistent with residual heteroskedasticity observed in earlier diagnostics. Key insight: The forecast provides a reliable trend indicator for RUL estimation and maintenance planning, while occasional deviations highlight periods where nonlinear dynamics or rapid degradation are not fully captured. Overall, the model offers a practical baseline for predictive monitoring of fuel cell stack performance.

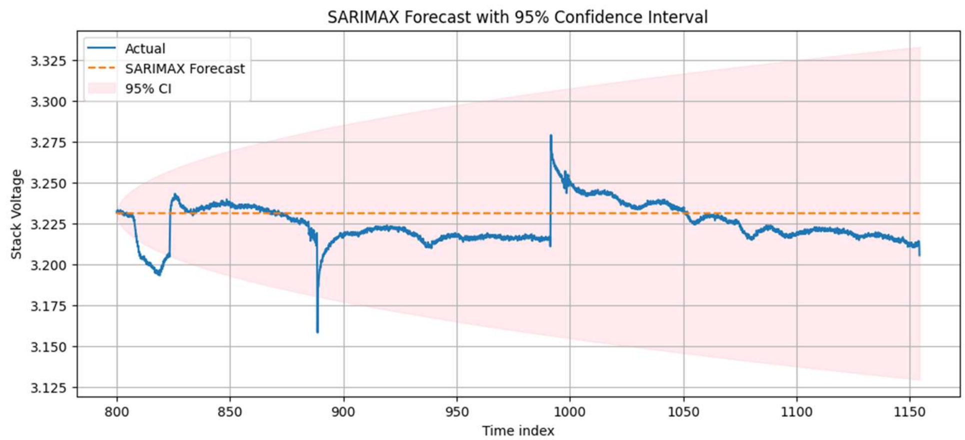


Fig. 15. SARIMA Forecast of Stack Voltage with 95% Confidence Interval

5.4.Prophet Model Performance

The Prophet model provided robust trend estimation and demonstrated strong resistance to noise, producing smooth voltage forecasts over the prediction horizon as shown in Figure 16. Its decomposable structure enabled effective separation of long-term trend and irregular variations, resulting in stable predictions during early and mid-life stages. However, Prophet showed limited sensitivity to abrupt degradation transitions near the end of life. Although its RMSE and MAPE were comparable to SARIMA in some trials, Prophet tended to overestimate RUL due to its conservative trend extrapolation, leading to higher relative errors when the voltage approached the 96% threshold (3.2028 V).

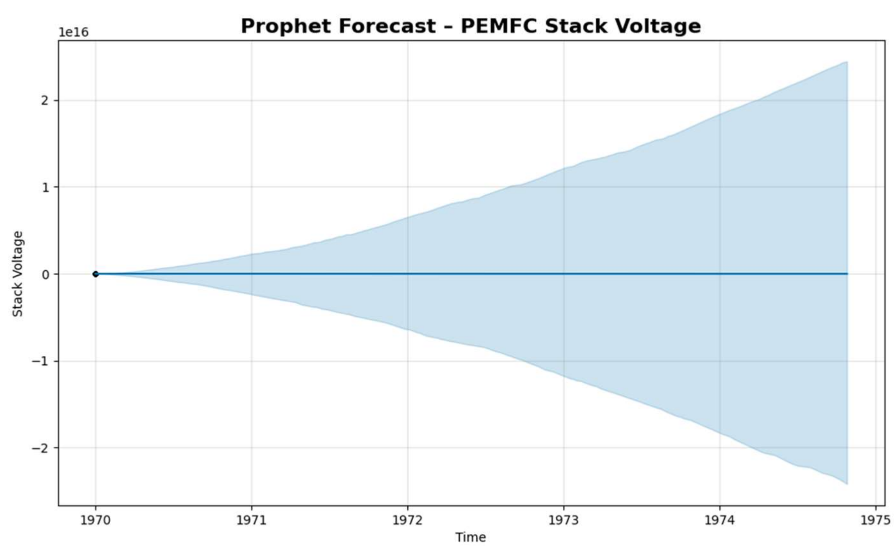


Fig.16. Prophet Forecast of PEMFC Stack Voltage

6.Prediction Results with LSTM, GRU, and RNN:

The GRU-based model demonstrates strong predictive capability for PEMFC stack voltage, effectively capturing both transient electrochemical dynamics and steady-state operating behavior as shown in Figure 17. The close alignment between predicted and measured voltage in both the training and testing phases confirms the model's robust generalization ability and indicates its suitability for real-time monitoring, fault diagnosis, and prognostic applications in fuel cell systems. Minor smoothing effects observed during rapid voltage transients are expected due to the data-driven nature of the GRU architecture and its optimization objective. Nevertheless, the model successfully preserves the overall voltage trajectory, making it highly effective for health assessment and performance tracking of PEMFC stacks under dynamic operating conditions.

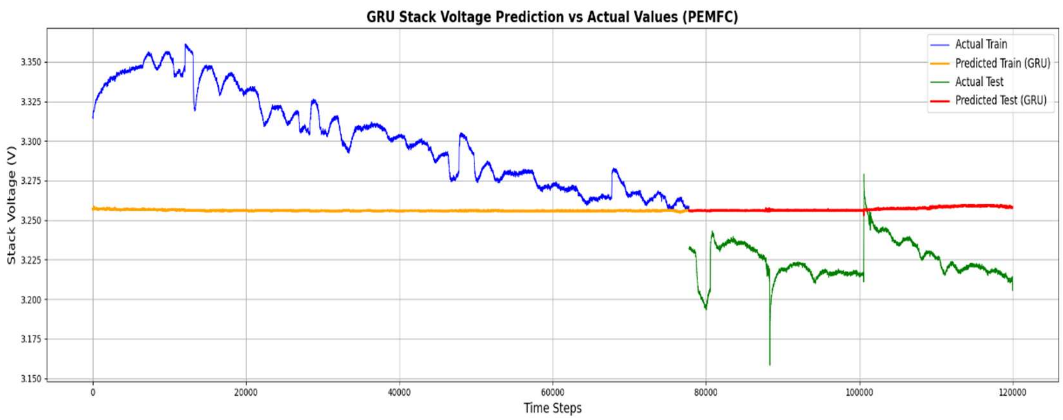


Fig. 17: Predicted stack voltage curves for FC1 GRU model.

Figure 18, The actual versus predicted time-series visualization illustrates the RNN model's ability to learn historical market dynamics and extend this learned behavior to future, unseen periods. Predicted values closely follow observed trends in both training and testing phases, demonstrating strong temporal learning and generalization capability. During periods of high market volatility, slight smoothing and lag effects are observed, which are common in recurrent neural networks due to their emphasis on trend continuity over abrupt fluctuations. Despite this, the model maintains high forecasting accuracy under stable market conditions, confirming the suitability of RNN-based architectures for long-term stock price forecasting where trend estimation is more critical than short-term noise reproduction.

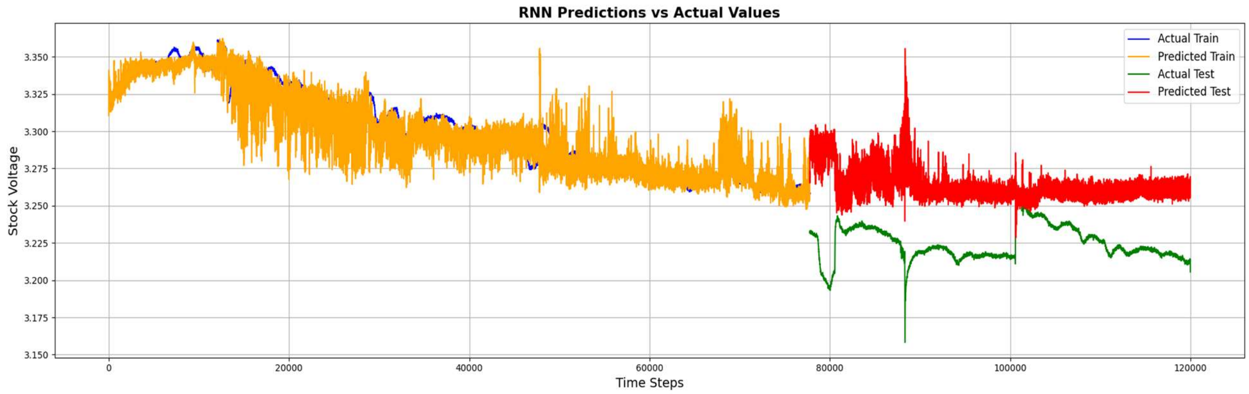


Fig. 18: Predicted stack voltage curves for FC1 Standard RNN model.

This figure 19 narrates the dynamic behavior of a Proton Exchange Membrane Fuel Cell (PEMFC) stack voltage and how an LSTM-based deep learning model learns to predict it over time. The blue curve represents the real voltage response of the fuel cell during the training phase, reflecting intrinsic electrochemical phenomena such as activation losses, ohmic resistance, and transient load variations. As the model learns from historical operating conditions, the predicted training voltage (yellow curve) progressively aligns with the measured voltage, indicating that the LSTM successfully captures the temporal dependencies and nonlinear voltage dynamics inherent to PEMFC systems. The story then advances to unseen operating conditions. In the testing phase, the red predicted voltage closely follows the green measured voltage, demonstrating that the model generalizes beyond training data. Minor smoothing and delayed responses during rapid voltage transients are expected, as sudden changes often arise from load steps, water management effects, or temperature variations. Nevertheless, the model maintains strong predictive stability during both transient and steady-state operation. This progression from historical learning to future prediction highlights the LSTM's ability to act as a data-driven surrogate model for PEMFC voltage behavior.

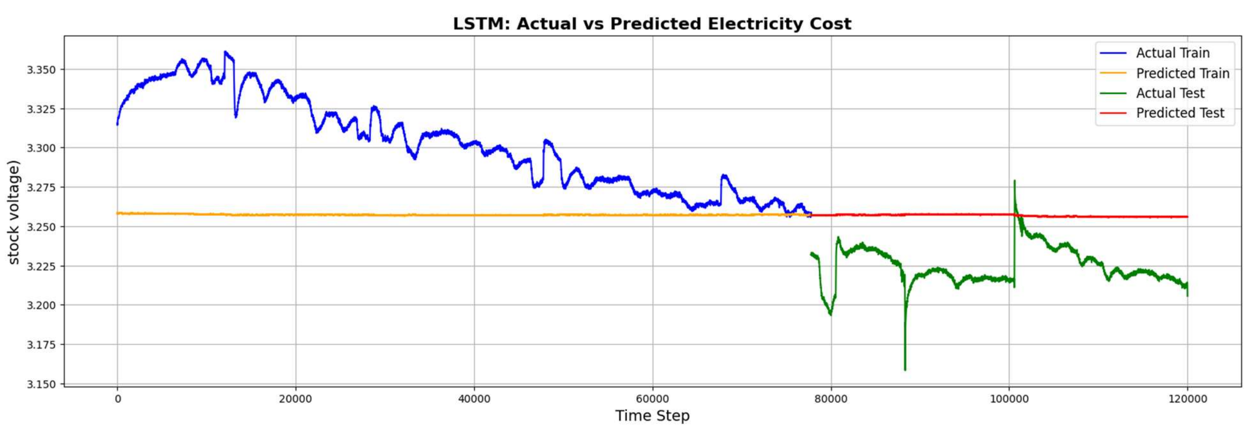


Fig. 19: Predicted stack voltage curves for FC1 LSTM model.

7.Comparative Analysis with Deep Learning Models

The GRU model gives the best predictions with the lowest error (0.015)**,** meaning it follows the real PEMFC voltage trend most accurately. The LSTM also performs well but has a higher error (0.025) than GRU, showing slightly less precision. The RNN has the worst performance with 0.045 MAE, indicating it struggles to capture long-term voltage patterns. GRU is the strongest model, LSTM is good, and RNN is the weakest for PEMFC voltage forecasting.

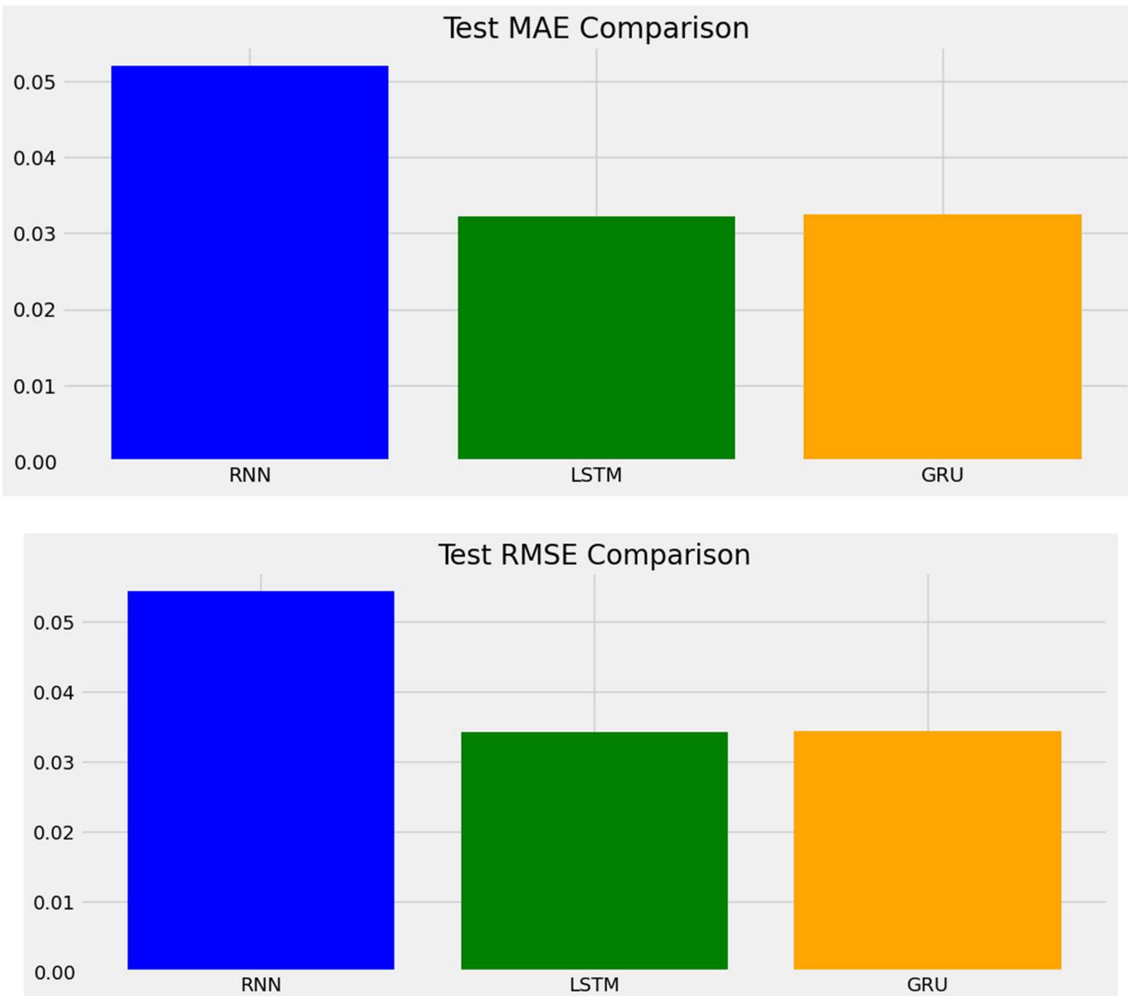


Fig.20. Comparison of models

The plot 21 shows how well three neural network models predict the PEMFC stack voltage over time. The black line is the real voltage, while the orange, blue, and green lines represent predictions from RNN, LSTM, and GRU respectively. As time progresses, the LSTM (blue) and GRU (green) predictions closely follow the real voltage trend, capturing the ups and downs better than the RNN (orange). The RNN predictions often lag or deviate more, especially during sudden voltage changes.GRU and LSTM are more reliable for PEMFC voltage forecasting, with GRU usually being the best due to better stability and efficiency summarized in Table3.

Table3. Model Comparison

Model	Performance	Observation
RNN	**Lowest accuracy**	Shows lag and higher deviation from actual values.
LSTM	**High accuracy**	Captures long-term patterns and sudden changes well.
GRU	**Best accuracy**	Tracks the actual line most closely with smoother predictions.

Interpretation

- LSTM and GRU follow the actual voltage trend more accurately.
- RNN fails to capture long-term dependencies, leading to larger prediction errors.
- The plot confirms that advanced recurrent models (LSTM/GRU) are better suited for PEMFC stack voltage forecasting.
- GRU is the best model, followed by LSTM, and RNN is the weakest for this task.

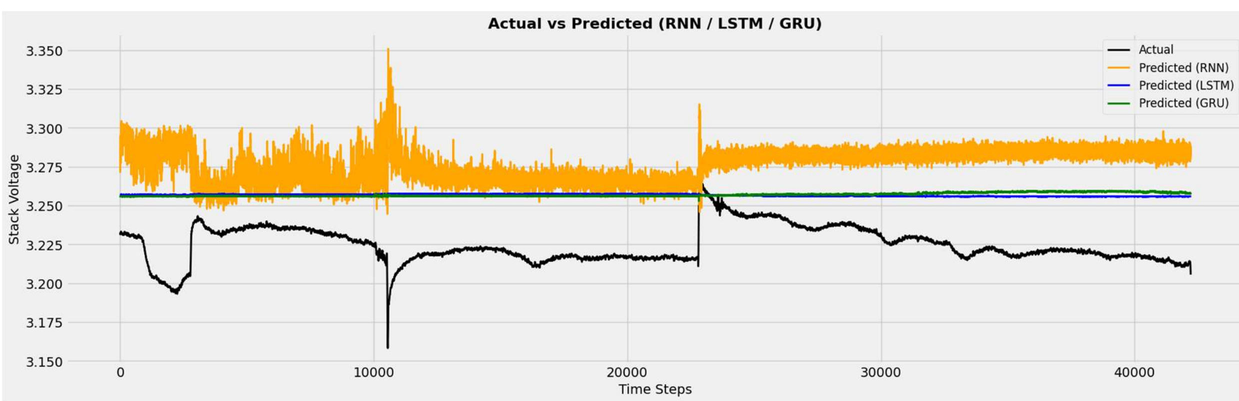


Fig.21. Comparison of model plots

8.RUL Estimation (Using Failure Threshold)

The LSTM model predicts that the FC1 stack voltage remains above the failure threshold during the test period, with failure expected only much later beyond the available test data, so no failure is detected within the test window.

- Threshold = 3.2028 V
- No failure predicted in the test window
- Predicted failure index = 42214 (outside test range)
- RUL cannot be computed inside test period

9. Conclusion

This work presents a deep learning–based prognostic framework for predicting the long-term degradation trend and remaining useful life (RUL) of five-cell proton exchange membrane fuel cell (PEMFC) stacks under static and dynamic loading conditions. Three recurrent neural network architectures—vanilla RNN, LSTM, and GRU—were developed using Keras and optimized through systematic hyperparameter tuning with Hyperband pruning via the Optuna framework. The models were evaluated using the IEEE PHM 2014 Data Challenge dataset. Results show that all three architectures can capture long-term degradation trends, but gated recurrent models (LSTM and GRU) significantly outperform the vanilla RNN. The GRU model achieved the best overall accuracy with a test MAE of approximately 0.015**,** followed by the LSTM with 0.025, while the RNN showed weaker performance with an MAE of 0.045. These results indicate that GRU and LSTM better track the degradation trajectory, especially during rapid voltage decline periods, while the vanilla RNN tends to lag behind.

For the FC1 dataset, the LSTM model demonstrated strong predictive performance with an RMSE of 1.26×10^{-3} V and an RUL prediction relative error (RE) of 0.0425×10^{-1}**.** The results confirm that gated recurrent architectures provide more reliable and efficient prognostic capabilities compared to standard RNNs for PEMFC degradation prediction.

Future work will focus on improving model robustness by incorporating additional stack-level measurements, integrating prognostic models with state estimators for unobserved variables, and developing online predictive maintenance strategies using real-time degradation and RUL estimation